Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used standard Python indentation (4 spaces) consistently throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project follows a modular structure with separate folders for templates, static files, model, preprocessing, and application logic. .
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as applicant_data, prediction_result, risk_score, and encoded_features clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as preprocess_input(), predict_approval(), and generate_audit_report() use meaningful names
5	Code Comments	Inline and block comments explain logic	Yes	Important sections and processing comme
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, prediction, feature alignment, and report generation are implemented as reusable modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Modular functions minimize code duplication and improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and

				exception handling are implemented to prevent application failures

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python (Pandas)	Input preprocessing and feature engineering	High
2	Prediction Engine	Python (Random Forest	Credit card approval prediction	High
3	Feature Alignment Module	Python (Scikit-learn)	Model input preparation	High
4	Prediction Logger	Python (JSON)	Stores prediction history and audit records	High
5	Risk Audit Report Generator	Python	Generates applicant risk assessment report	Medium

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Data Preprocessing Module	Python (Pandas)
2	Prediction Engine	Python (Random Forest)
3	Feature Alignment Module	Python (Scikit-learn)
4	Prediction Logger	Python (JSON)
5	Risk Audit Report Generator	Python